

TouchDesigner Operator-by-Operator Checklist (Spiral System)

This is the exact step-by-step build. Follow in order. No guessing.

STAGE 1 — PROJECT SETUP

1. Open TouchDesigner
 2. New project
 3. Set FPS: 60
 4. Create a Container COMP → rename to: `spiral_system`
 5. Enter the container (press `i`)
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STAGE 2 — OSC INPUT (REQUIRED FIRST)

Create nodes

1. TAB → CHOP → `OSC In` → rename: `oscin1`
2. Set Port: `11000`
3. TAB → CHOP → `Select` → rename: `select1`
4. Channels:

```
spiral/intensity  
spiral/filter_cutoff  
spiral/arp_density  
spiral/burst  
spiral/recovery  
spiral/brightness  
bio/heart_rate  
bio/breath_rate  
bio/stability
```

5. TAB → CHOP → `Rename` → rename: `rename1`
6. Map:

```
spiral/intensity → intensity
spiral/filter_cutoff → filter
spiral/arp_density → arp
spiral/burst → burst
spiral/recovery → recovery
spiral/brightness → brightness
bio/heart_rate → hr
bio/breath_rate → breath
bio/stability → stability
```

7. TAB → CHOP → **Math** → rename: **math_norm**

8. Normalize HR: 50-140 → 0-1

9. Normalize breath: 4-20 → 0-1

10. TAB → CHOP → **Null** → rename: **null_osc**

11. Turn ON "Cook Every Frame"

Connect

oscin1 → select1 → rename1 → math_norm → null_osc

STAGE 3 — CONTROL LAYER (SMOOTHING)

Create nodes

1. TAB → CHOP → **Lag** → rename: **lag1**

2. Lag: 0.05-0.12

3. TAB → CHOP → **Math** → rename: **math_control**

4. TAB → CHOP → **Null** → rename: **null_ctrl**

Connect

null_osc → lag1 → math_control → null_ctrl

STAGE 4 — SPIRAL CORE (FAST VERSION)

Create nodes

1. TAB → SOP → **Circle** → rename: **circle1**
2. Radius: 1
3. Divisions: 200
4. TAB → SOP → **Add** → rename: **add1**
5. Enable: "Polygons By Index"
6. TAB → SOP → **Transform** → rename: **xform_spiral**
7. TAB → COMP → **Geometry** → rename: **geo_core**
8. TAB → COMP → **Camera** → rename: **cam1**
9. TAB → COMP → **Light** → rename: **light1**
10. TAB → MAT → **Phong** → rename: **phong1**
11. TAB → TOP → **Render** → rename: **render_core**
12. TAB → TOP → **Null** → rename: **null_core**

Connect

circle1 → add1 → xform_spiral → geo_core geo_core → render_core → null_core

Assign: - geo_core → material: phong1 - render_core → camera: cam1 - render_core → light: light1

Add expressions

Rotation (Z):

```
op('null_ctrl')['arp'] * 200
```

Scale:

```
op('null_ctrl')['intensity'] * 0.8 + 0.4
```

STAGE 5 — PARTICLE HALO

Create nodes

1. TAB → SOP → **Sphere** → rename: **sphere1**
2. TAB → SOP → **Scatter** → rename: **scatter1**
3. Points: 200–2000
4. TAB → COMP → **Geometry** → rename: **geo_particles**
5. TAB → MAT → **Constant** → rename: **mat_particles**
6. TAB → TOP → **Render** → rename: **render_particles**
7. TAB → TOP → **Null** → rename: **null_particles**

Connect

sphere1 → scatter1 → geo_particles geo_particles → render_particles → null_particles

Assign material: - geo_particles → mat_particles

Expressions

Point count:

```
int(200 + op('null_ctrl')['brightness'] * 1800)
```

STAGE 6 — BURST FX

Create nodes

1. TAB → TOP → **Level** → rename: **level_burst**

2. TAB → TOP → Displace → rename: displace1

3. TAB → TOP → Transform → rename: transform_burst

4. TAB → TOP → Null → rename: null_burst

Connect

null_core → level_burst → displace1 → transform_burst → null_burst

Expressions

Brightness boost:

```
1 + op('null_ctrl')['burst'] * 2
```

Scale pulse:

```
1 + op('null_ctrl')['burst'] * 0.2
```

STAGE 7 — RECOVERY FX

Create nodes

1. TAB → TOP → Blur → rename: blur1

2. TAB → TOP → Level → rename: level_recovery

3. TAB → TOP → Null → rename: null_recovery

Connect

null_core → blur1 → level_recovery → null_recovery

Expression

Blur size:

```
op('null_ctrl')['recovery'] * 25
```

STAGE 8 — COMPOSITE

Create nodes

1. TAB → TOP → Composite → rename: comp1
2. TAB → TOP → Composite → rename: comp2
3. TAB → TOP → Bloom (or Glow) → rename: bloom1
4. TAB → TOP → Level → rename: level_final
5. TAB → TOP → Null → rename: null_final

Connect

null_core + null_particles → comp1 comp1 + null_burst → comp2 comp2 + null_recovery → bloom1 bloom1
→ level_final → null_final

Expression

Bloom strength:

```
op('null_ctrl')['brightness'] + op('null_ctrl')['burst'] * 0.3
```

STAGE 9 — UI DISPLAY

Create nodes

1. TAB → TOP → Text → rename: text_mode
2. TAB → TOP → Text → rename: text_hr
3. TAB → TOP → Layout → rename: layout1

4. TAB → TOP → Null → rename: null_ui

Connect

text_mode + text_hr → layout1 → null_ui

STAGE 10 — OUTPUT

Create nodes

1. TAB → TOP → Out → rename: out1

Connect

null_final → out1

FINAL TEST

1. Run Python system
2. Enable OSC
3. Confirm:
4. spiral rotates
5. brightness affects particles
6. burst flashes
7. recovery blurs

If all 4 work → system is alive

NEXT STEP

If you want next level: - I can build the **Python biofeedback module (mic + heart rate)** - or the **autonomous MIDI generator engine**